

F-02-CX-concentration-vs-Wplus-kill-test v1: does C_F^{conc} force $\int W_+ < \infty$? Quantitative audit

ZFL Lab

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MISSION FROZEN: Search only bridge $C_F^{conc} \Rightarrow \int_0^{T^*} W_+ < \infty$ or directly $C_E^{weak} \wedge C_{L^3} \wedge C_F^{conc} \Rightarrow \perp$. Attempt quantitative $W_+(t) \leq f(C_F^{conc}, \text{controlled})$ respecting NS identities under $T^* < \infty$.

INPUT: C_{PROVEN} pending audit, G LOCKED, E-CX CLOSED/BLOCKED, F-01=PROVEN NECESSARY NO CONTRADICTION if quantifiers correct. $W = \int \omega^T S \omega$, $W_+ = \int (\omega^T S \omega)_+$, $C_E^{weak} : \limsup \int_0^t W = +\infty \Rightarrow \int_0^{T^*} W_+ = \infty$, $C_{L^3} : \limsup \|u\|_3 = \infty$, $C_F^{conc} : \exists t_n \uparrow T^*, x_n, r_n \downarrow 0, \|u(t_n)\|_{L^3(B_{r_n}(x_n))} \geq \varepsilon_0$.

METHODOLOGICAL PRECISION: Compatibility killing must respect NS identities available under $T^* < \infty$: energy $\|u\|_{L_t^\infty L_x^2} \leq C$, enstrophy balance $\frac{1}{2}\|\omega\|_2^2 + \nu \int_0^t \|\nabla \omega\|_2^2 = \frac{1}{2}\|\omega_0\|_2^2 + \int_0^t W$, Calderon-Zygmund $\|S\|_p \sim \|\omega\|_p$, pressure $-\Delta p = \text{tr}(\nabla u)^2$, scaling $\|u^\lambda\|_3 = \|u\|_3$. Not abstract.

AUDIT:

Goal $W_+ \leq f(C_F^{conc})$.

Recall $|W| \leq \int |S||\omega|^2$. Hölder $|W| \leq \|S\|_{3/2} \|\omega\|_6^2$ and $\|\omega\|_6 \leq C \|\nabla \omega\|_2$. But $\|S\|_{3/2} \sim \|\omega\|_{3/2}$ not controlled by lower bound C_F^{conc} .

Critical: C_F^{conc} is LOWER bound $\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0$. W_+ needs UPPER bound. Lower cannot imply upper. No inequality $W_+ \leq f(\text{lower})$ holds.

Can C_F^{conc} coexist with arbitrarily large W_+ within NS? Yes: fix bubble with $\|u\|_{L^3(B_{r_n})} \geq \varepsilon_0$ and add high-frequency packet u_{high} with small $\|u_{high}\|_3$ (choose $M = \delta^{-1}$) that preserves lower bound but makes $W_+ \rightarrow \infty$ arbitrarily. Energy stays bounded, enstrophy balance allows large dissipation. Localized VST estimate shows factor $\sup \int |u|^2/R$ not a priori bounded without uniform L^3 bound which would imply regularity (ESS). Without it, VST can be large.

Thus within NS constraints, $C_F^{conc} + \int_0^{T^*} W_+ = +\infty$ compatible, arbitrary growth allowed.

KILL CONDITIONS CHECK:

- C_F^{conc} does not control any quantity entering W_+ (controls lower bound, W_+ needs upper)
- Only reproduces scale critical - Concentration can coexist with arbitrary W_+ within NS - Any bound would need extra hypothesis not consequence of $T^* < \infty$ - No integrated bound $\int W_+ < \infty$

VERDICT: $F-02 = KILLED$ fast

No bridge $C_F^{conc} \Rightarrow \int W_+ < \infty$, no bot. $F = CLOSED/BLOCKED \rightarrow H-01$ No F-03.

Pending: verify quantifiers in F-01 PDF for promotion to $\mathcal{C}_{PROVEN}$.